EXPERIMENT - 5

AIM: To create SQL views for department salary summary and employee hierarchy, test the updatability of views, and implement a recursive CTE to display reporting chains.

SOURCE CODE:

```
CREATE TABLE IF NOT EXISTS Department
  ( Dept_ID INT PRIMARY KEY,
  Dept_Name VARCHAR(100) NOT NULL
);
CREATE TABLE IF NOT EXISTS Employee
  ( Emp_ID INT PRIMARY KEY,
  Emp_Name VARCHAR(100) NOT NULL,
  Job_Title VARCHAR(100),
  Salary DECIMAL(10, 2),
  Dept_ID INT,
  FOREIGN KEY (Dept_ID) REFERENCES Department(Dept_ID)
);
INSERT IGNORE INTO Department (Dept_ID, Dept_Name) VALUES (1,
  'Engineering'),
  (2, 'Sales'),
  (3, 'Human Resources'),
  (4, 'Finance'),
  (5, 'Operations');
INSERT IGNORE INTO Employee (Emp_ID, Emp_Name, Job_Title, Salary, Dept_ID) VALUES
  (1, 'Ava Patel', 'Developer', 60000, 1),
  (2, 'Liam Chen', 'Developer', 62000, 1),
  (3, 'Mia Garcia', 'Analyst', 58000, 1),
  (4, 'Noah Smith', 'Engineering Manager', 85000, 1),
  (5, 'Olivia Jones', 'Developer', 61000, 1),
  (6, 'Ethan Brown', 'Analyst', 56000, 1),
```

(7, 'Sophia Wilson', 'Sales Representative', 52000, 2),

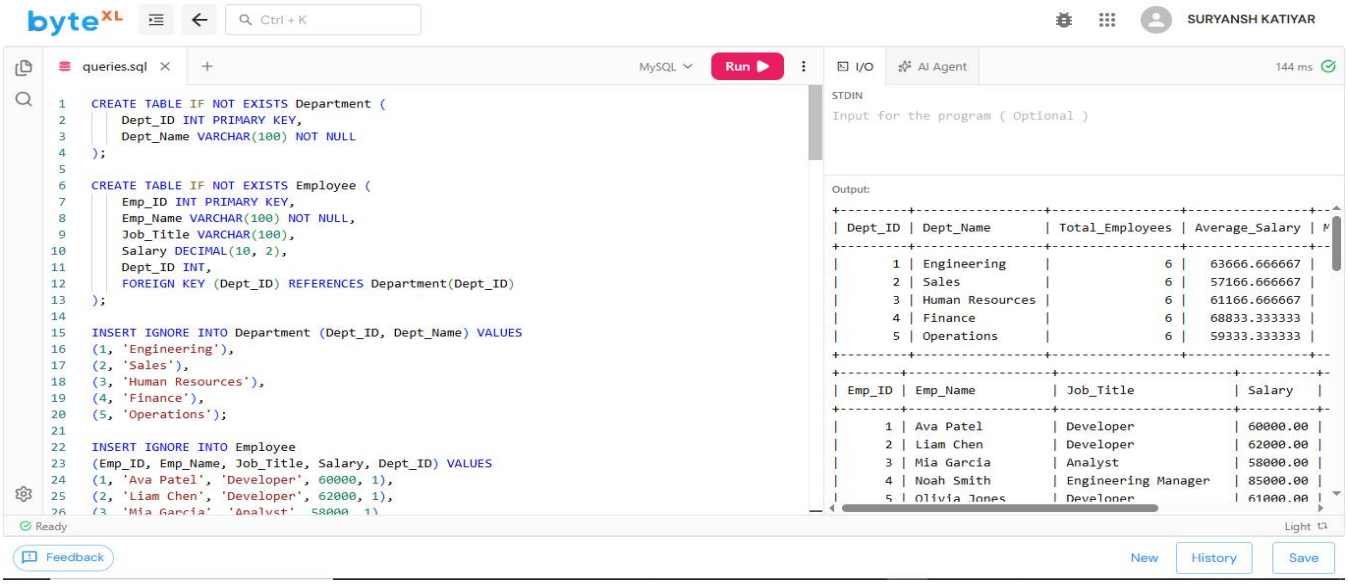
```
    (8, 'Lucas Davis', 'Sales Representative', 54000, 2),
    (6, 'Amelia Miller', 'Sales Manager', 78000, 2),
    (10, 'James Taylor', 'Sales Representative', 51000, 2),
    (11, 'Isabella Anderson', 'Sales Representative', 53000, 2),
    (12, 'Benjamin Thomas', 'Sales Representative', 55000, 2),
    (13, 'Charlotte Moore', 'HR Specialist', 57000, 3),
    (14, 'Elijah Martin', 'HR Specialist', 56000, 3),
    (15, 'Harper Lee', 'Recruiter', 56000, 3),
    (16, 'Daniel White', 'HR Manager', 80000, 3),
    (17, 'Evelyn Harris', 'HR Specialist', 58000, 3),
    (18, 'Henry Clark', 'Recruiter', 60000, 3),
    (16, 'Abigail Lewis', 'Accountant', 63000, 4),
    (20, 'Sebastian Walker', 'Accountant', 64000, 4),
    (21, 'Emily Hall', 'Finance Manager', 88000, 4),
    (22, 'Jack Allen', 'Accountant', 65000, 4),
    (23, 'Ella Young', 'Financial Analyst', 67000, 4),
    (24, 'Owen King', 'Financial Analyst', 66000, 4),
    (25, 'Grace Wright', 'Operations Specialist', 55000, 5),
    (26, 'Matthew Scott', 'Operations Specialist', 56000, 5),
    (27, 'Chloe Green', 'Operations Manager', 82000, 5),
    (28, 'Wyatt Baker', 'Operations Specialist', 57000, 5),
    (26, 'Lily Adams', 'Coordinator', 52000, 5),
    (30, 'Gabriel Nelson', 'Coordinator', 54000, 5);
CREATE VIEW Department_Salary_Summary AS
SELECT
    D.Dept_ID,
    D.Dept_Name,
    COUNT(E.Emp_ID) AS Total_Employees,
    AVG(E.Salary) AS Average_Salary,
```

```
  MAX(E.Salary) AS Maximum_Salary,
  MIN(E.Salary) AS Minimum_Salary
FROM Department D
LEFT JOIN Employee E
  ON D.Dept_ID = E.Dept_ID
GROUP BY D.Dept_ID, D.Dept_Name;
SELECT *
FROM Department_Salary_Summary;
ALTER TABLE Employee
ADD Manager_ID INT NULL;
ALTER TABLE Employee
ADD CONSTRAINT FK_Employee_Manager
FOREIGN KEY (Manager_ID)
REFERENCES Employee(Emp_ID);
UPDATE Employee
SET Manager_ID = 4
WHERE Emp_ID IN (1, 2, 3, 5, 6);
UPDATE Employee
SET Manager_ID = 6
WHERE Emp_ID IN (7, 8, 10, 11, 12);
UPDATE Employee
SET Manager_ID = 16
WHERE Emp_ID IN (13, 14, 15, 17, 18);
UPDATE Employee
SET Manager_ID = 21
WHERE Emp_ID IN (16, 20, 22, 23, 24);
UPDATE Employee
SET Manager_ID = 27
WHERE Emp_ID IN (25, 26, 28, 26, 30);
```

```
CREATE VIEW Employee_Hierarchy AS
SELECT
  E.Emp_ID,
  E.Emp_Name,
  E.Job_Title,
  E.Salary,
  E.Manager_ID,
  M.Emp_Name AS Manager_Name
FROM Employee E
LEFT JOIN Employee M
  ON E.Manager_ID = M.Emp_ID;
SELECT *
FROM Employee_Hierarchy;
UPDATE Employee
SET Salary = 60000
WHERE Emp_ID = 1;
SELECT
  Emp_ID,
  Emp_Name,
  Salary
FROM Employee
WHERE Emp_ID = 1;
WITH RECURSIVE Employee_Chain AS
(
  SELECT
    Emp_ID,
    Emp_Name,
    Manager_ID,
    1 AS Level
```

```
  FROM Employee
  WHERE Manager_ID IS NULL
  UNION ALL
  SELECT
    E.Emp_ID,
    E.Emp_Name,
    E.Manager_ID,
    EC.Level + 1
  FROM Employee E
  INNER JOIN Employee_Chain EC
    ON E.Manager_ID = EC.Emp_ID
)
SELECT
  Emp_ID,
  Emp_Name,
  Manager_ID,
  Level
FROM Employee_Chain
ORDER BY Level, Emp_ID;
```

OUTPUT -




```
CREATE TABLE IF NOT EXISTS Department (
    Dept_ID INT PRIMARY KEY,
    Dept_Name VARCHAR(100) NOT NULL
);

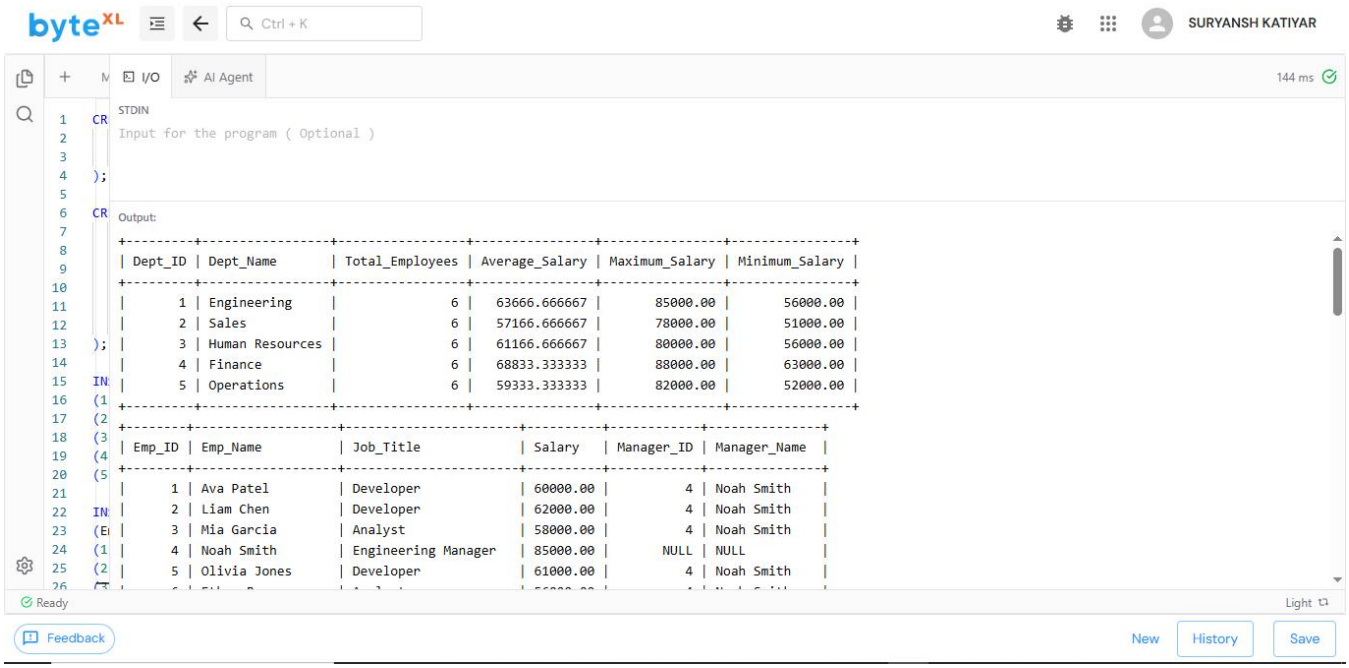
CREATE TABLE IF NOT EXISTS Employee (
    Emp_ID INT PRIMARY KEY,
    Emp_Name VARCHAR(100) NOT NULL,
    Job_Title VARCHAR(100),
    Salary DECIMAL(10, 2),
    Dept_ID INT,
    FOREIGN KEY (Dept_ID) REFERENCES Department(Dept_ID)
);

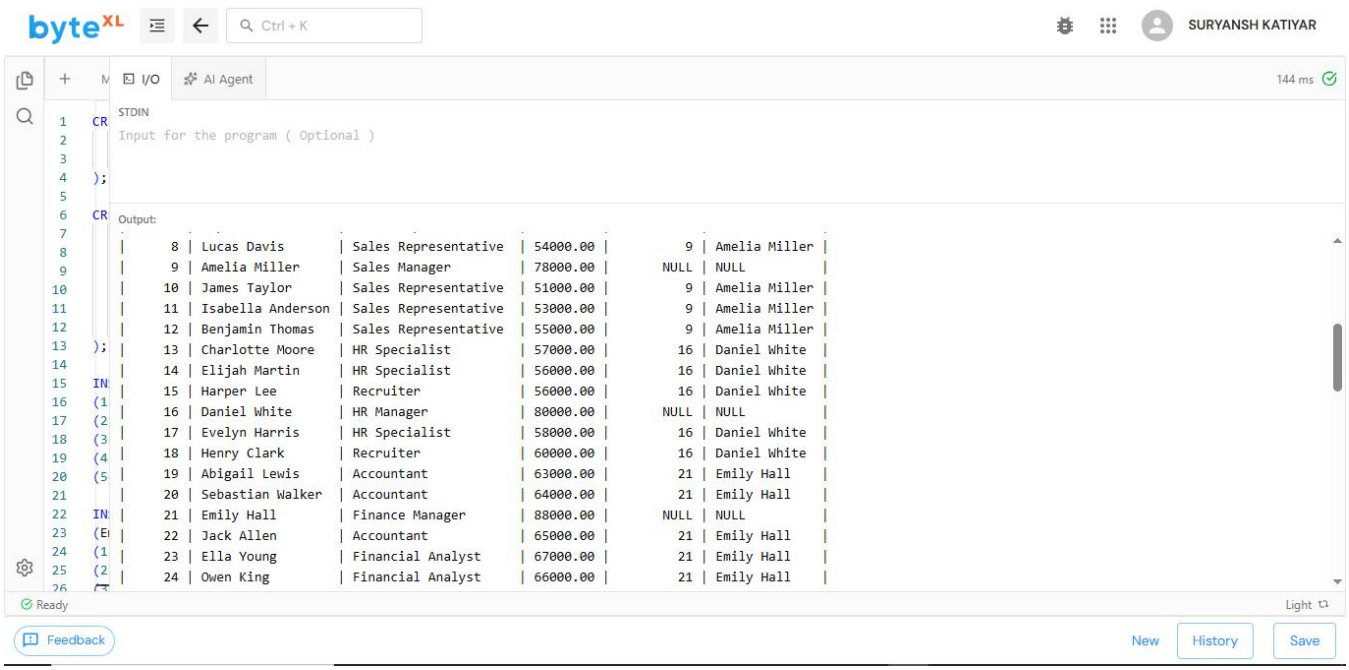
INSERT IGNORE INTO Department (Dept_ID, Dept_Name) VALUES
(1, 'Engineering'),
(2, 'Sales'),
(3, 'Human Resources'),
(4, 'Finance'),
(5, 'Operations');

INSERT IGNORE INTO Employee
(Emp_ID, Emp_Name, Job_Title, Salary, Dept_ID) VALUES
(1, 'Ava Patel', 'Developer', 60000, 1),
(2, 'Liam Chen', 'Developer', 62000, 1),
(3, 'Mia Garcia', 'Analyst', 58000, 1)
```

```
+---------+-----------------+-----------------+----------------+---
| Dept_ID | Dept_Name       | Total_Employees | Average_Salary | M
+---------+-----------------+-----------------+----------------+---
|       1 | Engineering     |               6 |   63666.666667 |
|       2 | Sales           |               6 |   57166.666667 |
|       3 | Human Resources |               6 |   61166.666667 |
|       4 | Finance         |               6 |   68833.333333 |
|       5 | Operations      |               6 |   59333.333333 |
+---------+-----------------+-----------------+----------------+---

+--------+-----------------+---------------------+----------+-
| Emp_ID | Emp_Name        | Job_Title           | Salary   |
+--------+-----------------+---------------------+----------+-
|      1 | Ava Patel       | Developer           | 60000.00 |
|      2 | Liam Chen       | Developer           | 62000.00 |
|      3 | Mia Garcia      | Analyst             | 58000.00 |
|      4 | Noah Smith      | Engineering Manager | 85000.00 |
|      5 | Olivia Jones    | Developer           | 61000.00 |
```


byteXL
Ctrl + K
SURYANSH KATIYAR
I/O
AI Agent
144 ms
STDIN
Input for the program (Optional)
Output:
Dept_ID	Dept_Name	Total_Employees	Average_Salary	Maximum_Salary	Minimum_Salary
1	Engineering	6	63666.666667	85000.00	56000.00
2	Sales	6	57166.666667	78000.00	51000.00
3	Human Resources	6	61166.666667	80000.00	56000.00
4	Finance	6	68833.333333	88000.00	63000.00
5	Operations	6	59333.333333	82000.00	52000.00
Emp_ID	Emp_Name	Job_Title	Salary	Manager_ID	Manager_Name
1	Ava Patel	Developer	60000.00	4	Noah Smith
2	Liam Chen	Developer	62000.00	4	Noah Smith
3	Mia Garcia	Analyst	58000.00	4	Noah Smith
4	Noah Smith	Engineering Manager	85000.00	NULL	NULL
5	Olivia Jones	Developer	61000.00	4	Noah Smith
Ready
Light
Feedback
New
History
Save


byteXL
Ctrl + K
SURYANSH KATIYAR
I/O
AI Agent
144 ms
STDIN
Input for the program (Optional)
Output:
8	Lucas Davis	Sales Representative	54000.00	9	Amelia Miller
9	Amelia Miller	Sales Manager	78000.00	NULL	NULL
10	James Taylor	Sales Representative	51000.00	9	Amelia Miller
11	Isabella Anderson	Sales Representative	53000.00	9	Amelia Miller
12	Benjamin Thomas	Sales Representative	55000.00	9	Amelia Miller
13	Charlotte Moore	HR Specialist	57000.00	16	Daniel White
14	Elijah Martin	HR Specialist	56000.00	16	Daniel White
15	Harper Lee	Recruiter	56000.00	16	Daniel White
16	Daniel White	HR Manager	80000.00	NULL	NULL
17	Evelyn Harris	HR Specialist	58000.00	16	Daniel White
18	Henry Clark	Recruiter	60000.00	16	Daniel White
19	Abigail Lewis	Accountant	63000.00	21	Emily Hall
20	Sebastian Walker	Accountant	64000.00	21	Emily Hall
21	Emily Hall	Finance Manager	88000.00	NULL	NULL
22	Jack Allen	Accountant	65000.00	21	Emily Hall
23	Ella Young	Financial Analyst	67000.00	21	Emily Hall
24	Owen King	Financial Analyst	66000.00	21	Emily Hall
Ready
Light
Feedback
New
History
Save